Ex:06

NMAP TO DISCOVER LIVE HOST USING NMAP SCANS (ARP, ICMP, TCP)

Aim:)

To Study the working of the Nmap tool and discover the host in a network using different host discovery techniques. Such as ARP scanning, DNS scanning, TCP scanning, and UDP on the given hardware platform.

Steps:

1. Open the given hardware platform and select the Nmap from
2. Study the network topology and identify the source and destination host
3. Use ARP request to discover the MAC address of device available in the local network
4. Observe how ARP Packets are broadcast with in the network and identify the devices that receives the request
5. Use the ICMP protocol to send ping request between 2 system and observed the responses
6. Analyses the packets flow and identify necessary ARP packets before ICMP communication
7. Perform ICMP Address Echo scanning and analyze the response generated by target System
8. Perform TCP ping scan and analyze the response generated by target System.

9. Perform UDP scanning and observe how Nmap identifies active hosts through UDP packets.

10. Perform a reverse DNS lookup to obtain the hostnames of the discovered systems.

11. Analyze the advantages and limitations of each scanning method.

Result:

Thus the live hosts networks were successfully identified using different Nmap scanning techniques. ARP was found to be effective for detecting devices in local networks, useful for identifying active